Experiment results appendix

This appendix traces manuscript results to saved source evidence. The original held-out experiment and exploratory extensions share the known 79-row holdout. Results do not imply prospective or external validation.

Held-out metrics

Model	Precision	Recall	F2	ROC-AUC	AP	Brier
Logistic regression	0.500	0.692	0.643	0.744	0.598	0.207
Decision tree	0.450	0.692	0.625	0.693	0.569	0.216
Random forest	0.583	0.538	0.547	0.723	0.676	0.209
Gradient boosting	0.647	0.423	0.455	0.721	0.564	0.197
Majority	0.000	0.000	0.000	0.500	0.329	0.329
Always-At-Risk	0.329	1.000	0.710	0.500	0.329	0.671
Prevalence probability	0.000	0.000	0.000	0.500	0.329	0.221
Prior failure	0.647	0.423	0.455	0.655	0.464	0.266

Confusion matrices and calibrated variants

Variant	TN	FP	FN	TP	Flags
LR	35	18	8	18	36
DT	31	22	8	18	40
RF	43	10	12	14	24
GB	47	6	15	11	17
LR__no_absences	35	18	9	17	35
DT__no_absences	28	25	6	20	45
LR__reduced_sensitive	39	14	12	14	28
DT__reduced_sensitive	43	10	15	11	21
LR__no_failures	28	25	7	19	44
DT__no_failures	22	31	4	22	53
LR__sigmoid	53	0	24	2	2
DT__sigmoid	53	0	24	2	2
majority	53	0	26	0	0
always_at_risk	0	53	0	26	79
prevalence	53	0	26	0	0
prior_failure	47	6	15	11	17

Evidence map

Table I: feature descriptions and source data dictionary. Tables II–III: heldout_metrics.csv and heldout_predictions.csv. Table IV: nested_summary.csv and nested_fold_metrics.csv. Table V: the same nested outputs grouped by feature subset. Table VI: heldout metrics for original and __sigmoid variants, with calibration_bins.csv. Table VII: subgroups.csv.

Figure 1: original confusion counts. Figure 2: saved ROC/precision-recall probabilities. Figure 3: dependent outer-fold F2 values. Figure 4: calibration bins and Wilson intervals. Figure 5: threshold_sensitivity.csv. Figure 6: subgroup recall and Wilson intervals. Figure 7: permutation_stability.csv. Files figure5_importance and figure6_thresholds are historical asset identifiers; visible manuscript numbering follows order of appearance.

Bootstrap evidence: bootstrap_intervals.csv, paired_differences.csv and mcnemar.json. Intervals condition on fixed fitted models and observed prevalence; no clustering or refitting uncertainty is covered. Feature schema: feature_subsets.json. Reproduction: reproduction_checks.json. Calibration protocol: protocol.json. Final audit: final/experiment_verification.json.

Recorded-sex subgroup denominators

Model/group	n	Failures	TP	FP	Negatives
LR/F	43	17	12	14	26
LR/M	36	9	6	4	27
DT/F	43	17	12	9	26
DT/M	36	9	6	13	27

Nested validation and retuned ablations

Each entry is mean (sample SD) across ten dependent folds, not a confidence interval.

Subset / model	F2 mean (SD)	AUC mean (SD)	AP mean (SD)
All 30 / Always risk	0.710 (0.006)	0.500 (0.000)	0.329 (0.007)
All 30 / DT	0.538 (0.113)	0.619 (0.044)	0.436 (0.042)
All 30 / GB	0.338 (0.094)	0.646 (0.048)	0.527 (0.068)
All 30 / LR	0.498 (0.140)	0.666 (0.091)	0.545 (0.114)
All 30 / Majority	0.000 (0.000)	0.500 (0.000)	0.329 (0.007)
All 30 / Prevalence	0.000 (0.000)	0.500 (0.000)	0.329 (0.007)
All 30 / Prior failure	0.422 (0.137)	0.639 (0.066)	0.448 (0.068)
All 30 / RF	0.468 (0.145)	0.665 (0.067)	0.526 (0.081)
No absences / DT	0.496 (0.094)	0.622 (0.027)	0.472 (0.067)
No absences / LR	0.503 (0.118)	0.659 (0.077)	0.536 (0.092)
No failures / DT	0.506 (0.092)	0.560 (0.076)	0.403 (0.091)
No failures / LR	0.525 (0.103)	0.610 (0.081)	0.473 (0.095)
Reduced 15 / DT	0.568 (0.123)	0.642 (0.072)	0.464 (0.046)
Reduced 15 / LR	0.530 (0.111)	0.678 (0.076)	0.570 (0.100)

Conditional held-out bootstrap intervals

2,000 class-stratified paired resamples, seed 2701; percentile 95% intervals. Fixed models and prevalence; no cluster or retraining uncertainty.

Model / metric	Point	Lower	Upper
LR / accuracy	0.6709	0.5696	0.7722
LR / precision	0.5000	0.3889	0.6250
LR / recall	0.6923	0.5000	0.8462
LR / f1	0.5806	0.4516	0.6909
LR / f2	0.6429	0.4861	0.7770
LR / AUC	0.7438	0.6284	0.8454
LR / AP	0.5985	0.4714	0.7560
LR / Brier	0.2072	0.1709	0.2473
DT / accuracy	0.6203	0.5190	0.7215
DT / precision	0.4500	0.3556	0.5556
DT / recall	0.6923	0.5000	0.8462
DT / f1	0.5455	0.4262	0.6552
DT / f2	0.6250	0.4744	0.7566
DT / AUC	0.6930	0.5609	0.8197
DT / AP	0.5688	0.4276	0.7321
DT / Brier	0.2164	0.1711	0.2624
RF / accuracy	0.7215	0.6203	0.8101
RF / precision	0.5833	0.4231	0.7619
RF / recall	0.5385	0.3462	0.7308
RF / f1	0.5600	0.4000	0.7059
RF / f2	0.5469	0.3659	0.7143
RF / AUC	0.7228	0.5914	0.8375
RF / AP	0.6763	0.5363	0.8053
RF / Brier	0.2091	0.1924	0.2265
GB / accuracy	0.7342	0.6456	0.8228
GB / precision	0.6471	0.4499	0.8571
GB / recall	0.4231	0.2308	0.6154

Model / metric	Point	Lower	Upper
GB / f1	0.5116	0.3181	0.6809
GB / f2	0.4545	0.2609	0.6400
GB / AUC	0.7213	0.6016	0.8309
GB / AP	0.5637	0.4335	0.7331
GB / Brier	0.1968	0.1547	0.2404

Full subgroup denominators

Rates use observed positives for recall and observed negatives for false-positive rate.

Model / attribute / group	n / positives	Recall	FPR
LR / sex / F	43 / 17	0.706	0.538
LR / sex / M	36 / 9	0.667	0.148
LR / address / R	20 / 7	0.857	0.692
LR / address / U	59 / 19	0.632	0.225
LR / school / GP	69 / 23	0.696	0.326
LR / school / MS	10 / 3	0.667	0.429
DT / sex / F	43 / 17	0.706	0.346
DT / sex / M	36 / 9	0.667	0.481
DT / address / R	20 / 7	0.857	0.538
DT / address / U	59 / 19	0.632	0.375
DT / school / GP	69 / 23	0.696	0.413
DT / school / MS	10 / 3	0.667	0.429
RF / sex / F	43 / 17	0.471	0.231
RF / sex / M	36 / 9	0.667	0.148
RF / address / R	20 / 7	0.857	0.231
RF / address / U	59 / 19	0.421	0.175
RF / school / GP	69 / 23	0.522	0.174
RF / school / MS	10 / 3	0.667	0.286
GB / sex / F	43 / 17	0.294	0.154

Model / attribute / group	n / positives	Recall	FPR
GB / sex / M	36 / 9	0.667	0.074
GB / address / R	20 / 7	0.714	0.154
GB / address / U	59 / 19	0.316	0.100
GB / school / GP	69 / 23	0.391	0.109
GB / school / MS	10 / 3	0.667	0.143